Algorithms in computing exercises

01-E01 Ordering, preservation, and stability

Sort the records [(2,a),(1,b),(2,c),(1,d)] by nondecreasing key, stably. Identify the specifications violated by [(1,b),(1,d),(2,c),(2,a)] and [(1,b),(2,a),(2,c)]. Give a fully correct output. Define the result for empty input.

01-E02 First match and termination

Trace FIRST-MATCH on [8,3,8,5] for 8 and for 9. List every key comparison and return value. Give the loop invariant and termination variant. Explain short-circuit evaluation and handle $n = 0$.

01-E03 Partial correctness is insufficient

Legal inputs are all nonnegative integers n. A program returns 0 when n is even and otherwise repeatedly performs a harmless operation forever. The required output is 0. Prove partial correctness, give a counterexample to total correctness, and give a totally correct replacement.

01-E04 Paths and tours

On the complete undirected graph with vertices 0,1,2,3, perimeter edges 01,12,23,30 have weight 1 and diagonals 02,13 have weight 2. Find a shortest 0-to-2 path and a minimum TSP tour. Fix start 0, identify reversals, and list the three candidate tours and their costs. State what this enumeration proves and what it does not prove.

01-E05 Conditions for randomized majority

For every fixed legal input, a binary-decision algorithm has error probability at most $1/4$ per run and terminates within 1 s. Run it independently three times and use majority vote. Bound error and time. Explain what fails if runs share exactly the same randomness or if outputs are legal paths with several possible correct answers.

01-E06 Integer threshold in one model

For $n \geq 2$, let $A(n) = 2n^2$ steps and $B(n) = 40n \lg n$ steps at the same processing rate. Find the smallest N such that $B < A$ for every integer $n \geq N$. Prove no later reversal and check $N - 1$. If the formulas are also imposed at $n = 1$, why is the smallest isolated solution different from the persistent threshold?

Algorithms in computing exercises (continued)

01-E07 Input bits and hardware units

Machine A performs $2n^2$ steps at 10^8 steps/s. Machine B performs $40n \lg n$ steps at 10^6 steps/s. Compute both times at $n = 1024$. A separate program performs N unit steps on a binary integer N. What is its worst-case order for an L-bit input? Check the length and count for $N = 31$.

01-E08 Parallelism is not a processor cutoff

Six independent unit tasks must all finish before a chain of three unit tasks starts. Find work, span, minimum completion times, and speedups for $P = 1, 2, 3, 6$. Use two of these processor counts to refute the claim that $P > W/S$ prevents further benefit.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study